



Separation of antibody monomer-dimer mixtures by frontal analysis



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ABSTRACT

The removal of aggregates, particularly soluble dimers, from monoclonal antibodies (mAbs) remains a persistent challenge in downstream processing. In this work, we have examined the separation of an antibody monomer from its dimer on the cation exchange resin Nuvia HR-S (Bio-Rad Laboratories) using frontal analysis. In this process, a mixture of monomer and dimer is continuously fed to the column under conditions where the mixture is favorably bound, resulting in two breakthrough fronts whose monomer and dimer compositions are determined by the multi-component equilibrium and kinetics of the system. Experimentally, the selectivity for dimer was found to vary substantially with ionic strength, being lowest when conditions favor the strongest binding, and increasing to a maximum at intermediate ionic strengths where rapid exchange with the bound monomer can occur. A mechanistic model is developed to describe the competitive binding frontal analysis process, assuming pore diffusion and a significant kinetic resistance to binding as a function of ionic strength. The model was solved numerically and was able to describe both the frontal analysis processes and batch adsorption experimental data, accounting for process parameters such as feed composition and salt concentration. The resulting model can be used to optimize column operating conditions for yield and purity.

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1. Introduction

Removal of aggregates from therapeutic antibodies is a critical step in biopharmaceutical manufacturing as a result of both safety and efficacy concerns [1,2]. Chromatography is most commonly used for this purpose. While the removal of larger aggregates is usually easily achieved, removal of soluble dimers often presents a challenging chromatographic separation since the selectivity for monomer-dimer separation is often low. Additionally, mass transfer is slow, thereby requiring long columns or very long residence times to achieve separation. Cation exchange (CEX) chromatography is used frequently in these applications since many antibodies (and their dimers) have a relatively high pI [3–5]. Although, in theory, conditions may be found where the monomer flows through while the dimer binds strongly, for robustness reasons, the separation is more frequently conducted in a bind and elute mode. In this case, the column is loaded with the protein mixture to be separated, and then eluted either isocratically or with a gradient in either salt concentration or pH [6,7]. Since the separation results from differential migration of monomer and dimer along the gra-

dient, protein loading is typically limited to a fraction of the column binding capacity. In turn, these protein load limitations reduce the productivity. In view of increasing demands to improve downstream process productivity [8,9], there is an interest in exploring alternative chromatographic modalities.

In our previous work [10], we investigated the adsorption equilibrium and kinetics of a monoclonal antibody (mAb) and its soluble dimer in a macroporous cation exchange resin. Both binding equilibrium behavior and binding kinetics were investigated for monomer and dimer at both high and low protein loadings, and both individually and for the two-component system. We found that conditions could be identified where (a) selective elution of the two species occurred in a salt gradient at low protein loads, and (b) displacement of one species by the other occurred over relatively short time scales at high protein loads. In general, effective displacement of the monomer by the dimer required intermediate salt concentrations. Almost no exchange between the two species was observed over reasonable time scales at low salt concentration, when binding is very strong. However, at intermediate salt concentrations, favorable exchange at rates that were nearly entirely limited by diffusional transport was observed.

Although our prior work was carried out exclusively in batch mode, the results led to the conclusion that it might be possible to exploit the monomer-dimer exchange process to obtain sepa-

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ration by frontal analysis. In such a process, the monomer-dimer feed mixture would be continuously loaded to the column. Ideally, the more weakly bound species (the monomer in our case) would emerge from the column first, concentrated in a pure band as a result of competition with the dimer for binding sites. Then, immediately prior to dimer breakthrough, the column would be washed to recover the unbound monomer and eluted at high salt to remove the dimer along with any co-adsorbed monomer.

Compared to bind-elute processes, a potential advantage of frontal analysis is that the column binding capacity is fully utilized, which, in turn, can lead to greater productivity. Its success depends, however, on having both selectivity and sufficiently fast kinetics so that the competitive binding process can occur over reasonable time scales. Since predicting purity and yield in such a process is more complicated, a model capable of describing the column dynamics is desirable.

Protein separations by frontal analysis have been studied by prior authors for a few different systems. Weinbrenner and Etzel [11] demonstrated the separation of BSA monomer and dimer by frontal loading on CEX membranes. Fargues et al. [12] showed the competitive breakthrough of a system of albumin and hemoglobin on hydroxyapatite supports. Garke et al. [13] showed evidence of competitive binding of γ -globulin/lysozyme mixtures on a CEX column. These authors also developed a modified Langmuir model to describe competitive binding in batch adsorption, but did not provide the extension of the model to predict the column behavior. Hunter and Carta [14] examined frontal analysis of BSA and BSA aggregate mixtures on AEX chromatography columns. Xu and Lenhoff [15] also developed a model, based on colloidal energetics, to predict binary protein adsorption on CEX resins. While successful, the model was only tested against relatively small model proteins such as lysozyme, cytochrome C, and ribonuclease. A few studies have also recently been published on binary adsorption of antibodies. Liu et al. [16] explored the frontal loading of mAbs on CEX columns, quantifying the breakthrough of aggregates and other impurities over a variety of stationary phases and loading conditions, but did not provide a mechanistic model to describe the separation. Tao et al. [17] studied the separation of mAb charge variants by frontal analysis and provided a mechanistic model assuming full mass transfer control, but did not consider the separation of aggregates.

This work has two primary objectives. The first is to investigate experimentally the frontal analysis separation of an antibody monomer-dimer system at varying salt concentration and residence times. The second objective is to develop a mechanistic model to describe the separation, with the goal of being able to predict experimental conditions that provide an optimum balance of yield and purity. The model parameters for the proteins are based on our previously reported experimental data for protein binding on the macroporous CEX resin Nuvia HR-S, which were obtained based both on macroscopic measurements and on microscopic measurements using confocal microscopy [10].

2. Experimental

2.1. Materials

The cation exchanger used in this work is Nuvia HR-S (Bio-Rad Laboratories, Hercules, CA, USA), a polymeric resin with a macroporous structure and apparent pore radius of approximately 55 nm as determined by inverse size exclusion chromatography (iSEC) using neutral dextran probes (see Fig. S1a in Supplementary information). The volume-average particle diameter is 52.6 with a range of $\pm 5.7 \mu\text{m}$, as determined from micrographs (see Fig. S1b in Supplementary information). This resin was chosen because of

its inertness toward catalyzing the on-column formation of aggregates compared to other resins [18], relatively large pore size to allow diffusion of the monomer and dimer with comparatively low hindrance, and small particle size to provide relatively fast mass transfer.

The protein samples used in this work were the same as those used previously [10], and were provided by MedImmune (Gaithersburg, MD, USA). The samples included a high purity IgG1 antibody containing low levels of higher molecular weight species (>97% monomer by SEC), and a sample of the same IgG1 containing a high percentage of soluble aggregates (<68% monomer by SEC). The latter sample was obtained as the aggregate-rich fraction from a hydroxyapatite chromatography step following low-pH elution from a Protein A column. All other chemicals and salts were purchased from Fisher-Scientific (Waltham, MA, USA) and Sigma-Aldrich (St. Louis, MO, USA). All experiments were performed at room temperature ($22 \pm 2^\circ\text{C}$).

2.2. Methods

2.2.1. Sample preparation and analytics

Frozen protein samples were thawed at room temperature, and buffer exchanged using a HiPrep 26/10 Desalting column (GE Healthcare, Piscataway, NJ, USA) equilibrated with 10 mM Na_2HPO_4 at pH 7.0 with varying amounts of NaCl added. Preparative size exclusion chromatography (SEC) was used to isolate the antibody dimer from the aggregate-rich feedstock with a Superdex 200 Increase 10/300 GL column (GE Healthcare, Piscataway, NJ, USA). The separation was run at 0.5 mL/min in 50 mM Na_2HPO_4 , 150 mM NaCl, pH 6.8 with 0.5 mL samples of approximately 5 mg/mL serially injected. The resulting dimer peaks were collected, concentrated with a pressurized Amicon® stirred-cell module using a 30 kDa regenerated cellulose UF membrane from Millipore (Billerica, MA, USA), and then buffer-exchanged as described above.

Protein concentrations were measured using a NanoDrop 2000c UV-vis spectrophotometer (Thermo Scientific, Wilmington, DE, USA) at 280 nm. A mass-based extinction coefficient of $1.33 \text{ mL mg}^{-1} \text{ cm}^{-1}$ (determined from a protein assay) was used to quantify both monomer and dimer solutions by converting UV absorbance to concentration.

The composition of solutions containing monomer-dimer mixtures was determined by UPLC-SEC using a Waters Acquity H-Class UPLC unit (Milford, MA, USA) with an Acquity UPLC Protein BEH SEC column (200 Å pore, 1.7 μm particles). The mobile phase was 100 mM Na_2HPO_4 , 300 mM NaCl, pH 6.8 and the flow rate was 0.3 mL/min, which resulted in an analysis time of about 7 min with a monomer-dimer resolution of >1.5. The injection volume was 5 μL . The ratio of peak areas was used to calculate percentage of monomer and dimer, and the total area was used to determine the protein concentration based on a calibration curve obtained with the protein feedstock used in each experiment. It should be noted that the SEC column efficiency changed somewhat over time as the column aged and was replaced when the monomer-dimer resolution fell below 1.

2.2.2. Frontal analysis experiments

Frontal analysis experiments were conducted using laboratory scale glass columns 0.5 cm in diameter (a Tricorn 5/100 with column packing extension tube and a HR 5/5 column were used, both obtained from GE Healthcare, Piscataway, NJ, USA). The columns were flow packed with the resin to varying bed heights between 3 and 11 cm by first preparing an approximately 50% slurry in pH 7 phosphate buffer with 1 M NaCl added, pouring a calibrated volume of the slurry in the column, quickly inserting the top adapter, and then running the mobile phase (pH 7 phosphate with 1 M NaCl added) at 2.5 mL/min until the bed was consolidated. An ÄKTA

Table 1
Equilibrium binding capacities obtained from the single and binary breakthrough experiments compared with predictions from the SMA model.

[Na ⁺] (mM)	$C_{M,F}$ (mg/mL)	$C_{D,F}$ (mg/mL)	$C_{M,1}/q_{M,1}^*(C_{M,1})$ (mg/mL)		$q_{M,1}^*(C_{M,F}, C_{D,F})$ (mg/mL)		$q_{D,1}^*(C_{M,F}, C_{D,F})$ (mg/mL)	
			BT curve	SMA model	BT curve	SMA model	BT curve	SMA model
20	2.0	0	–	–	130	140	–	–
20	0	2.1	–	–	–	–	140	150
20	1.6	0.73	2.0/130	2.3/140	91	8.9	71	140
30	1.4	0.52	1.6/110	1.8/130	70	8.8	83	130
50	1.3	0.52	1.8/110	1.8/100	25	7.8	96	110
80	2.0	0	–	–	76	74	–	–
80	0	2.0	–	–	–	–	99	95
80	1.4	0.52	1.8/77	1.9/73	9	7	87	83
120	1.4	0.64	1.7/30	1.8/36	5	5	67	53

Explorer 10 system (GE Healthcare, Piscataway, NJ, USA) was used for this purpose. Packing quality was tested by measuring the HETP for a pulse injection of salt using the ÄKTA system conductivity probe. Reduced HETP values and asymmetry factors for salt were 6.0 ± 1.2 and 0.97 ± 0.15 , respectively. The extraparticle (or external) void fractions of these columns were obtained from the retention of pulse injections of blue dextran, assuming that it is completely excluded from the resin pores, and were $\varepsilon = 0.38 \pm 0.03$. The load flow rate was adjusted between 0.25 and 0.5 mL/min dependent on column length to give residence times between 2.4 and 4.6 min.

Single and two-component breakthrough experiments were performed on an ÄKTA Explorer 10 (GE Healthcare, Piscataway, NJ, USA) with a Frac-950 fraction collector. All protein feeds were prepared with a total protein concentration of 2 ± 0.3 mg/mL, and loaded onto the column for 120–380 column volumes depending on the load buffer Na⁺ concentration and feed composition. Actual feed compositions are given in Table 1. The effluent protein concentration was monitored by UV at 280 nm. For the two-component experiments, fractions were also collected in 4 CV increments and analyzed off-line via UPLC-SEC. After loading was completed, the column was eluted with 1 M NaCl, and the resulting eluate pool was also analyzed via UPLC-SEC. Protein recoveries in the elution pool were in excess of 90%.

3. Results and discussion

3.1. Experimental results

Fig. 1a and b show the single-component breakthrough curves for monomer and dimer, respectively, in buffers containing 20 and 80 mM Na⁺. For both monomer and dimer, breakthrough occurs earlier at 80 mM Na⁺ because of the lower binding capacity. The dimer breakthrough curves are significantly shallower than those obtained for the monomer, a result consistent with the lower pore diffusivity of the dimer determined in our prior work using confocal microscopy [10]. The figures also show model predictions, which are discussed in subsequent sections.

Fig. 2 shows representative binary breakthrough curves. Results are shown for load buffers containing 20, 50, 80 and 120 mM Na⁺ with a feed containing approximately 2 mg/mL total protein with 70% monomer. In addition to the monomer and dimer breakthrough profiles, the monomer purity, calculated as $C_M/(C_M + C_D)$, is also shown. Binary breakthrough results for load buffers containing 30 mM Na⁺ were also obtained and are included in the Supplementary information (Fig. S2). The area between monomer and dimer curves is directly related to the selectivity. As seen in Fig. 2a, little separation occurs at 20 mM Na⁺. However, the separation improves dramatically at higher Na⁺ concentrations with a well-defined band of purified monomer. For example, with 80 mM Na⁺ (Fig. 2c), fractions with monomer purity greater than 95% are obtained at the column outlet between approximately 25 and

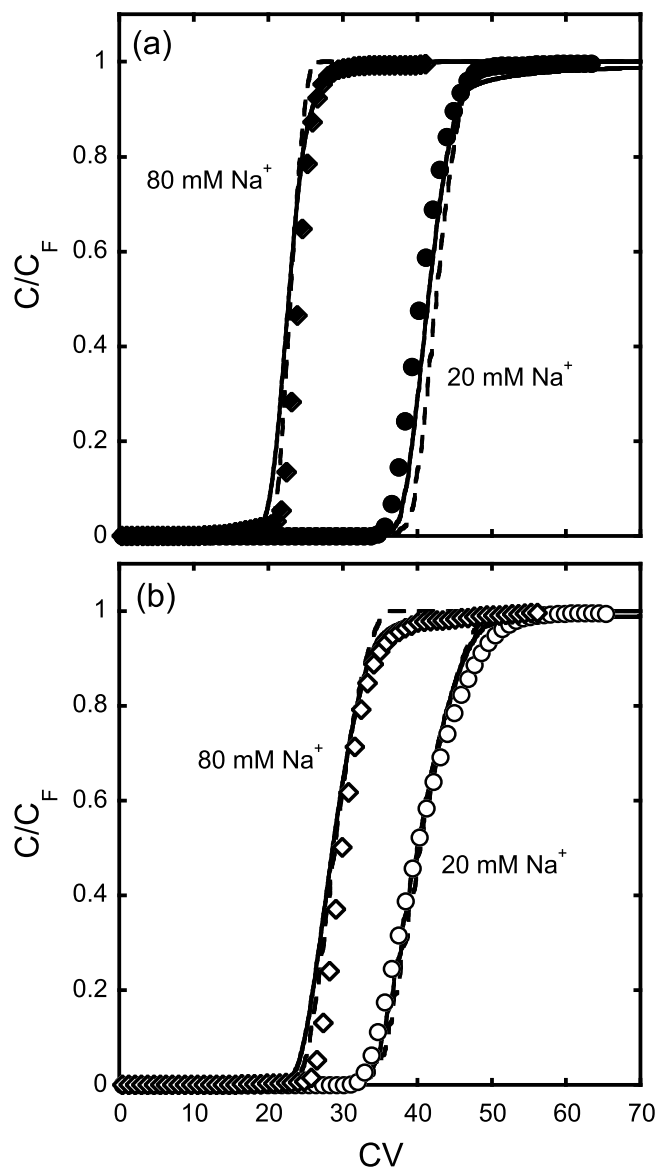


Fig. 1. One-component breakthrough curves of (a) monomer and (b) dimer in pH 7 load buffers containing 20 and 80 mM Na⁺. Feed concentration = 2.0 ± 0.1 mg/mL, $u = 76$ cm/h, column length = 4.4 cm, residence time = 3.5 min. Dashed lines are model predictions neglecting any kinetic resistance to binding. Solid lines are fitted including a kinetic resistance to binding with rate constants given in Table 2.

75 CV, corresponding to feed loads between 50 and 150 mg total protein per mL of column. Each of the experimental binary breakthrough curves exhibits an overshoot of the monomer front above the feed concentration ($C_M/C_{M,F} > 1$), which confirms that the

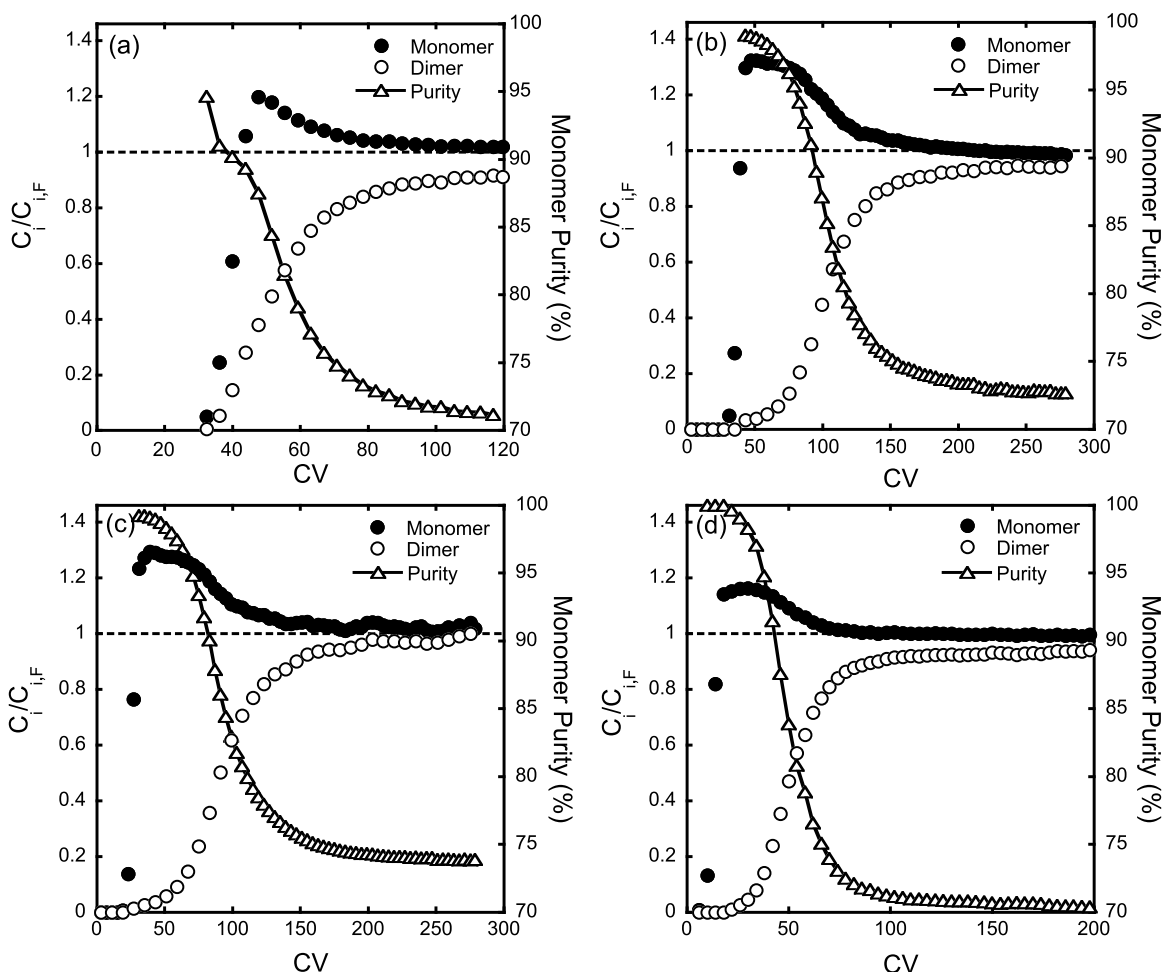


Fig. 2. Binary breakthrough behavior of monomer-dimer mixtures containing approximately 30% dimer in pH 7 load buffers containing different Na^+ concentrations. (a) 20 mM, (b) 50 mM, (c) 80 mM, (d) 120 mM. u was 150 cm/h in (a) and 76 cm/h in (b), (c), and (d); column lengths were 10.6 cm in (a), 5.9 in (b), 5.0 in (c), and 5.6 in (d); residence times were 4.2 min in (a), 4.6 in (b), 3.9 in (c), and 4.4 in (d). Exact feed compositions are given in Table 1.

dimer displaces the bound monomer. The magnitude and duration of the overshoot is, however, dependent on the Na^+ concentration, ranging from a modest overshoot of very brief duration at 20 mM Na^+ , when binding is strongest and selectivity is apparently smallest, to a well defined overshoot at 50, 80 and 120 mM Na^+ , when binding is weaker but selectivity is higher. The results obtained with 30 mM Na^+ given in Fig. S2 in Supplementary information follow a consistent trend with respect to separation.

Fig. 3 shows the SEC results for the eluate pools, obtained by eluting the column with 1 M NaCl following each of the breakthrough runs in Fig. 2, in comparison with those of the particular feed mixture used. Consistent with the breakthrough curves, at 20 mM Na^+ the eluate pool chromatogram is very similar to that of the feed. Conversely, much less monomer is found in the eluate pool compared to the feed for the 50 mM Na^+ run and almost no monomer is found in the eluate pool at 80 and 120 mM Na^+ . It is evident, however, that in addition to the dimer, some higher order species are also present in the eluate pool, as demonstrated by the early eluting peak in the SEC chromatograms. We believe that these species are present as impurities in the original dimer sample and are concentrated on the resin during the frontal analysis process.

3.2. Binding capacities

The monomer and dimer equilibrium binding capacities were obtained directly by integrating the breakthrough curves. Under

local equilibrium conditions, the pure component breakthrough curve would consist of a shock emerging from the column at:

$$CV_{sh} = \varepsilon + (1 - \varepsilon) \frac{q^*(C_F)}{C_F} \quad (1)$$

where ε is the extraparticle (or external) porosity, $q^*(C_F)$ is the bound protein concentration in equilibrium with the feed (in units of mg per mL of particle volume), C_F is the protein feed concentration, and CV represents the effluent volume divided by the column volume [19,20]. Accordingly, the equilibrium binding capacities can be calculated from the area above each breakthrough curve from the following equation:

$$q^*(C_F) = \frac{\int_0^\infty (C_F - C) dCV}{(1 - \varepsilon)} - \varepsilon C_F \quad (2)$$

where C is the actual effluent protein concentration. For the binary feed, again under local equilibrium conditions and assuming that the dimer can displace the monomer, the breakthrough curve would comprise two shocks with an intermediate plateau of pure monomer at a concentration $C_{M,1}$. These two shocks would occur at $CV_{sh,1}$ and $CV_{sh,2}$ given by [19,20]:

$$CV_{sh,1} = \varepsilon + (1 - \varepsilon) \frac{q_M^*(C_{M,1}, 0)}{C_{M,1}} \quad (3)$$

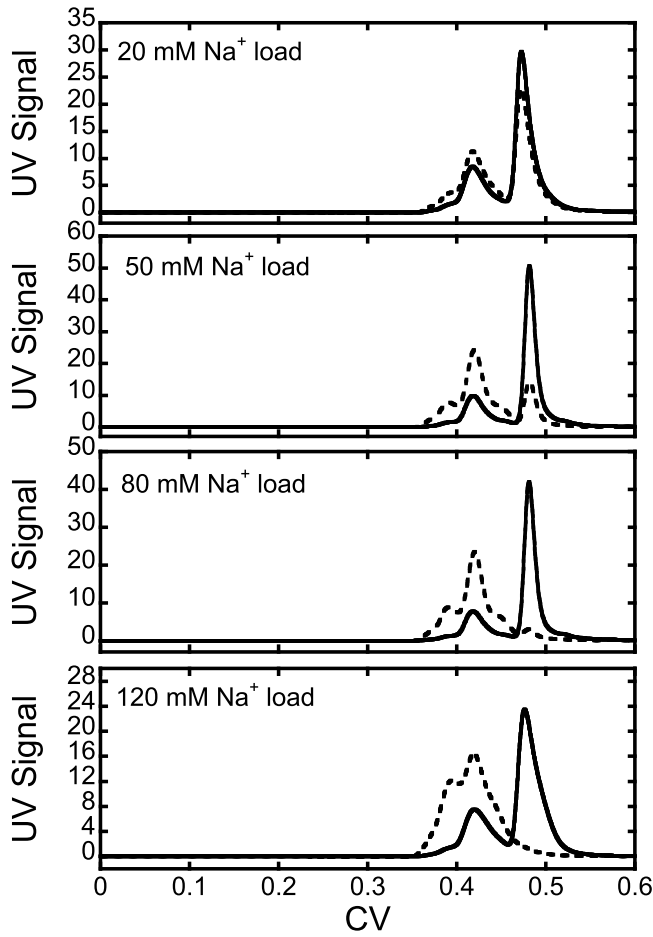


Fig. 3. SEC analyses of the eluate pools corresponding to the frontal analysis runs in Fig. 2. The dashed lines show the chromatograms of the protein pool eluted in 1 M NaCl. Solid lines show the chromatograms of each feed mixture.

$$CV_{sh,2} = \varepsilon + (1 - \varepsilon) \frac{q_M^*(C_{M,F}, C_{D,F}) - q_M^*(C_{M,1}, 0)}{C_{M,F} - C_{M,1}}$$

$$= \varepsilon + (1 - \varepsilon) \frac{q_D^*(C_{M,F}, C_{D,F})}{C_{D,F}} \quad (4)$$

where $C_{M,F}$ and $C_{D,F}$ are the monomer and dimer feed concentrations. As seen in Fig. 2, the experimental profiles are spread out and only roughly approximate the local equilibrium shocks. Nevertheless, the values of $q_M^*(C_{M,1}, 0)$, $q_M^*(C_{M,F}, C_{D,F})$, and $q_D^*(C_{M,F}, C_{D,F})$ can still be calculated approximately from Eqs. (3) and (4) by substituting the centers of mass of the breakthrough curves of pure monomer and dimer in place of $CV_{sh,1}$ and $CV_{sh,2}$ as was done for Eq. (2).

The results of these calculations are shown in Table 1 where they are compared with predictions based on single component isotherm measurements using the steric mass action (SMA) law model [21]:

$$q_i^* = \frac{K_{e,i} [q_0 - (z_M + \sigma_M) q_M^* - (z_D + \sigma_D) q_D^*]^{z_i}}{[Na^+]^{z_i}} C_i \quad (5)$$

where q_0 is the resin charge density, K_e is the equilibrium constant, z_i is the effective protein binding charge, and σ_i is the steric hindrance parameter. The parameters appearing in this equation, q_0 , K_e , z_i , and σ_i , were those obtained in our prior work [10] from a

global fit of single component isotherms over different Na^+ concentrations.

As seen in Table 1, the overshoot concentrations, $C_{M,1}$, and the corresponding q^* -values derived from the pure monomer front are in reasonable agreement with the SMA isotherm values at all Na^+ concentrations. However, agreement for the mixture q^* -values, derived from the monomer/dimer front, is highly dependent on the Na^+ concentration. At 20 mM Na^+ , when protein binding is strongest, there is very poor agreement with the SMA model. On the other hand, at higher Na^+ concentrations, when protein binding is weaker, the agreement improves significantly and becomes quantitative for both monomer and dimer at 80 and 120 mM Na^+ . This result is consistent with the conclusions reached in our previous work [10], where, based on batch isotherm data we demonstrated that the SMA model was only suitable at relatively high Na^+ concentrations. In fact, based on Eq. (5), the monomer-dimer selectivity, $\alpha_{D,M} = q_D^* C_M / q_M^* C_D$, is predicted to be inversely proportional to $[Na^+]^{z_D - z_M}$. Since the dimer effective charge, z_D , is greater than that of the monomer, z_M , the SMA model predicts lower selectivity for the dimer at higher Na^+ concentration. This trend is observed experimentally at relatively high Na^+ , but not at all at low Na^+ concentrations. Results obtained with 30 mM Na^+ are given in Fig. S2 in Supplementary information and follow a consistent trend with respect to binding capacities.

3.3. Breakthrough modeling

The model used to describe the breakthrough curves for both single and two-component systems is based on two main assumptions. The first is that plug flow is assumed with axial dispersion simulated through the numerical dispersion introduced by a backwards-finite difference discretization of the relevant material balance equation. The second is that intraparticle transport is assumed to be controlled by pore-diffusion. The first of these assumptions is based on the fact that the reduced velocity, $v' = v d_p / D_0$, where v is the interstitial velocity, d_p is the particle diameter, and D_0 is the protein solution diffusivity, is well above 500 for the conditions studied. In this case, mass transfer, rather than axial dispersion, dominates band broadening so that an exact description of axial dispersion is unnecessary [22]. The second assumption is based on our prior work which, based on confocal microscopy imaging, showed that one-component adsorption at low Na^+ concentrations follows very closely the shrinking core model [10]. The effective pore diffusivities $D_e = 8.0 \times 10^{-8}$ and 4.1×10^{-8} cm²/s for the mAb monomer and mAb dimer, respectively, were determined, and are, as expected, 5–10 times lower than the corresponding free solution diffusivities as a result of diffusional hindrance. External mass transfer is also neglected based on the large value of the Biot number, $Bi = k_f r_p / D_e$, where k_f is the boundary layer mass transfer coefficient and r_p is the particle radius. Using k_f -values estimated from engineering correlations [23] gave Biot numbers well in excess of 40, demonstrating that the external resistance is negligible. Finally, although the one-component adsorption kinetics is expected to be fast, a kinetic resistance to binding can be expected for the binary case since this involves the competitive binding of molecules with large differences in size and binding charge. Accordingly, our model also includes a kinetic contribution to binding, which is represented by the kinetic expression of the SMA model.

The model equations along with initial and boundary conditions are as follows:

$$\varepsilon \frac{\partial C_i}{\partial t} - \frac{3(1 - \varepsilon) D_{e,i}}{r_p} \frac{\partial C_i}{\partial r} \bigg|_{r=r_p} + u \frac{\partial C_i}{\partial x} = 0 \quad (6)$$

$$C_i|_{t=0} = 0, C_i|_{x=0} = C_{i,F} \quad (6a)$$

Table 2
Parameters used in model calculations.

General Parameters		
Column length	See figure captions	
Superficial velocity, u	See figure captions	
Particle radius, r_p (μm)	26	
Column extraparticle porosity, ε	0.38	
Resin intraparticle porosity, ε_p	0.50	
Individual Protein Parameters		
	Monomer	Dimer
D_e (cm^2/s) ^a	8.0×10^{-8}	4.1×10^{-8}
q_0 (mM) ^a	132	
z^a	10.2	14.8
σ^a	111	229
K_e^a	104	1.33×10^4
Rate Parameters for Eq. (8)		
$[\text{Na}^+]$ (mM)	k_M ($\text{mM}^{-z} \text{s}^{-1}$)	k_D ($\text{mM}^{-z} \text{s}^{-1}$)
20	1×10^{-17}	1.3×10^{-23}
30	3×10^{-18}	1×10^{-20}
50	3×10^{-20}	1×10^{-24}
80	1×10^{-21}	4×10^{-29}
120	2×10^{-23}	3×10^{-31}

^a From Ref. [10].

$$\varepsilon_p \frac{\partial c_i}{\partial t} + \frac{\partial q_i}{\partial t} = \frac{D_{e,i}}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial c_i}{\partial r} \right) \quad (7)$$

$$c_i|_{t=0} = 0, \quad \frac{\partial c_i}{\partial r} \Big|_{r=0} = 0, \quad c_i|_{r=r_p} = C_i \quad (7a)$$

$$\frac{\partial q_i}{\partial t} = k_i \left\{ c_i \left[q_0 - \sum_j (z_j + \sigma_j) q_j \right]^{z_i} - \frac{q_i [\text{Na}^+]^{z_i}}{K_{e,i}} \right\} \quad (8)$$

$$q_i|_{t=0} = 0 \quad (8a)$$

In these equations, C_i and c_i are the protein concentrations in the mobile phase and within the particle pores, respectively, u is the mobile phase superficial velocity, x is the column axial coordinate, q_i is the local bound protein concentration at a radial position r within the particle, ε_p is the intraparticle porosity, and k_i is a rate constant accounting for the finite rate of exchange between protein and Na^+ counterions. The model equations (Eqs. (6)–(8)) were solved numerically for both one and two-component systems by spatially discretizing Eqs. (6) and (7) by finite differences and integrating the resulting system of time-dependent ODEs using the *ode15s* variable order differential equation solver in MATLAB R2015a (The Mathworks, Natick, MA, USA). The SMA model parameters were obtained from single-component isotherm data reported in our prior work [10]. Additional model parameters (e.g. ε , u , r_p) were set to match experimental conditions and measured resin properties. The rate constant values, k_M and k_D , for monomer and dimer, respectively, were fitted to the experimental breakthrough profiles. The model parameters are summarized in Table 2.

Model predictions of the one-component breakthrough curves are given in Fig. 1. Two cases are shown, one without a kinetic resistance to binding, i.e. assuming a completely mass-transfer control process (dashed lines) and one including a kinetic resistance as described by Eq. (8) (solid lines). For the mass transfer controlled case, the values of k_M and k_D were chosen to be sufficiently high that further increasing them did not change the shape of the model predicted breakthrough curves. In this scenario, the concentrations at each local discretization point in the numerical solution scheme are essentially under equilibrium conditions as described by the multi-component SMA isotherm of Eq. (5). Conversely, for the case where the kinetic resistance is included, the values of k_M and k_D were adjusted to minimize the difference between experimental and model profiles. As seen in this figure, the difference between

the two cases, with and without kinetic resistance to binding, is very small, suggesting that the exchange between either monomer or dimer for Na^+ counterions on the CEX surface is fast.

Model predictions for the binary breakthrough curves are shown in Fig. 4. As before, the dashed lines are predictions without a kinetic resistance while the solid lines include this resistance according to Eq. (8). For the cases of 20 and 80 mM Na^+ , Fig. 4a and c, respectively, the values of the kinetic constants k_M and k_D are the same as those obtained by fitting the corresponding one-component breakthrough curves. For the other two cases (Fig. 4b and d), k_M and k_D were adjusted to provide the best possible agreement with the experimental curves. As seen in Fig. 4a, whether a kinetic resistance is included or not, model predictions deviate grossly from the experimental profiles. Without a kinetic resistance, the predicted separation is much better than seen experimentally, whereas including the kinetic resistance, the model predicted separation is much worse. This occurs because, as discussed above, the SMA model fails to predict the two-component equilibrium for these very strong binding conditions. In this case, no set of k_M and k_D values results in predicted breakthrough curves that follow the experimental ones. Agreement could only be obtained if the equilibrium model matched the experimental q^* values. On the other hand, as seen in Fig. 4b–d, higher Na^+ concentrations lead to model predicted curves that are much closer to the experimental ones. Moreover, it can be seen that the difference between curves calculated with and without a kinetic resistance become much closer to each other as the Na^+ concentration is increased. In turn, this suggests that as the Na^+ concentration increases, the frontal analysis process becomes increasingly mass-transfer controlled and, thus, predictable without including the kinetic resistance to binding. Results obtained with 30 mM Na^+ are given in Fig. S2 in Supplementary information and follow the same trends with respect to modeling.

A final consideration is whether the model developed in this work can (a) predict the effect of load composition and (b) provide results that are consistent with our prior observation of intraparticle concentration profiles during co-adsorption of monomer-dimer mixtures. The first consideration is of obvious practical value, while the second is important to confirm the physical validity of the model that we have developed as, usually, agreement with single particle-level measurements provides a stricter test of the validity of the model assumptions. Fig. 5 compares model predictions with the experimental two-component breakthrough curves at 80 mM Na^+ obtained with a feed containing 85% monomer. Additionally, the experiment was run on a different length column with a significantly shorter residence time than the previous 80 mM Na^+ experiment. The experimental SEC chromatograms of the feed and eluate pool are also shown (Fig. 5b). As seen in Fig. 5a, the model prediction (adjusting only the feed concentrations and column length) is in excellent agreement with the experimental concentration and purity profiles. In this case, since the initial monomer purity is higher, purity levels above 95% are obtained between about 30 and 130 CV, corresponding to the range 60–260 mg total protein loaded per mL of column volume. Note that, in agreement with the model, the monomer concentration overshoot is smaller than in the previous cases where the dimer content of the feed was higher. SEC analysis of the eluate pool (Fig. 5b) shows that at the end of the loading process the resin contains primarily bound dimer with a certain amount of higher order aggregates that were concentrated in the column as a result of their stronger binding.

Fig. 6 compares model predictions with the intraparticle concentration profiles obtained by confocal microscopy for the co-adsorption of a 2.2 mg/mL monomer-dimer mixture containing 85% monomer at pH 7 in a buffer containing 80 mM Na^+ . The confocal images obtained at three different times (4, 40, and 160 min), reported in our prior paper [10] and included in Fig. 6, were

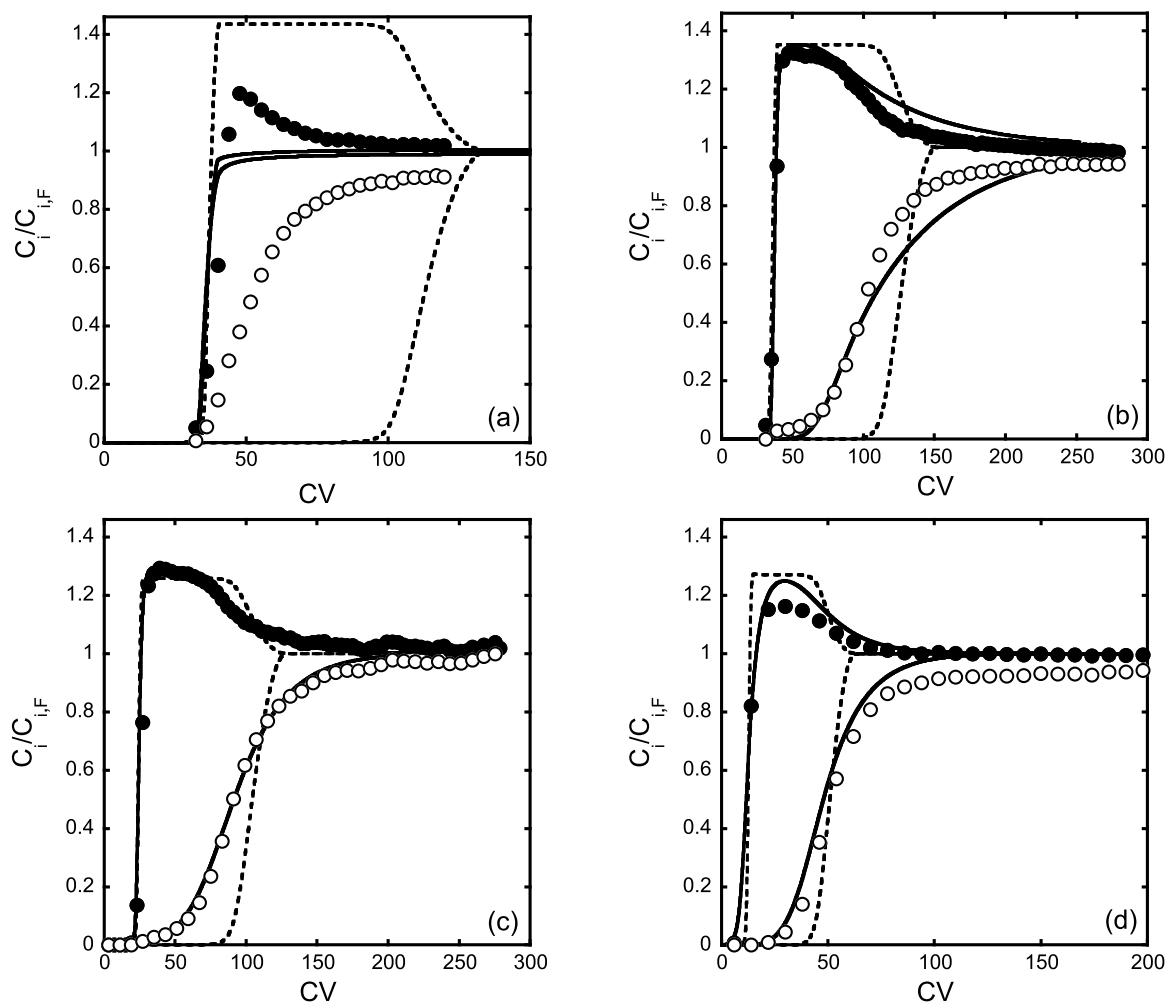


Fig. 4. Comparison of binary breakthrough curves with predictions based on Eqs. (6)–(8) using the SMA model (Eq. (5)) to describe competitive binding of monomer and dimer. Experimental conditions are the same as in Fig. 2. Dashed lines show predictions without including a kinetic resistance to binding. Solid lines include a kinetic resistance to binding according to Eq. (8) with rate constants given in Table 2.

obtained using monomer fluorescently labeled with Rhodamine Red dye and dimer fluorescently labeled with Rhodamine Green dye, thereby allowing the simultaneous imaging of each component during adsorption of their mixture. The particles imaged have diameters of $55 \pm 1 \mu\text{m}$, which is comparable to the mean particle size. The radial intraparticle concentration profiles were obtained from the gray-scale intensities using ImageJ 1.42q image analysis software [24] assuming that red and green fluorescence intensities were proportional to the bound protein concentrations. Since the exact relationship between fluorescence intensity and protein concentration is unknown and varies with the microscope settings, the intensity curves were normalized so that their radial average would match the model-predicted profiles. As seen in Fig. 6, the model predicts nearly quantitatively the shape of the experimental intensity profiles, further confirming its validity. In agreement with the data, the model predicts relatively fast kinetics with relatively diffuse intraparticle concentration profiles, with the more weakly bound monomer rapidly advancing toward the center of the particle ahead of the dimer early on (Fig. 6a), followed by much slower rates later (Fig. 6b and c), with the dimer gradually displacing the bound monomer. It is notable that even after 160 min, a substantial amount of monomer remains bound as a result of the relatively slow kinetics of competitive binding. This feature is clearly captured by our model.

4. Conclusions

The ability to conduct mAb monomer-dimer separations by frontal analysis with a CEX resin was demonstrated for practical conditions. This required selecting buffer compositions where the binding strength is intermediate, thereby facilitating displacement of the monomer by the dimer. For these conditions, the binary breakthrough curves exhibit all of the characteristics expected for frontal analysis; namely, a pure monomer front reaching concentrations above the feed value, followed by a monomer-dimer front reaching the feed concentrations. Adsorption capacities determined from material balances show that the two-front behaviors are quantitatively consistent with predictions from the SMA equilibrium model only at relatively high Na^+ concentrations. Poor agreement is found at low Na^+ , as for these conditions the SMA model predicts selectivity trends that are opposite of those observed experimentally. Fortunately, these are also conditions where the selectivity is low and thus practically unimportant.

The mechanistic model that we have developed is limited to conditions where the SMA isotherm provides an adequate description of equilibrium; namely, at relatively high Na^+ concentrations. The model has predictive ability, but also provides insight about the factors that control the kinetics of the process. Comparison of model predictions and experimental results show that while intraparticle diffusion is the dominant contribution to band broadening,

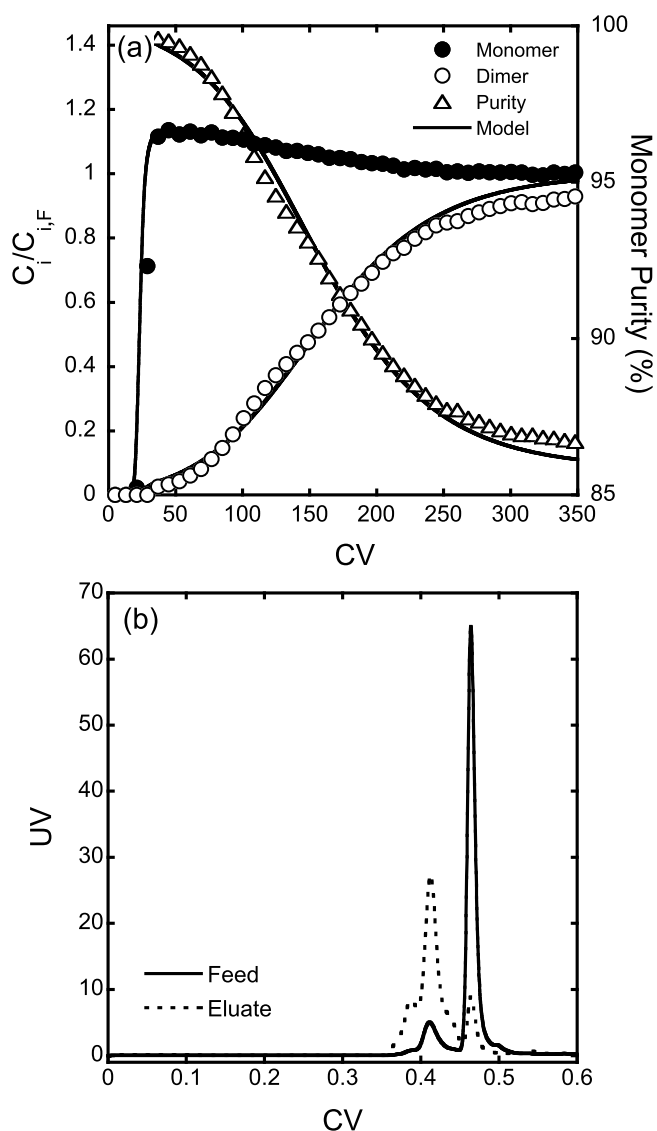


Fig. 5. Binary breakthrough behavior of a 2 mg/mL monomer-dimer mixture containing 15% dimer in a pH 7 load buffer with 80 mM Na⁺. (a) breakthrough curves, (b) SEC analysis of feed and 1 M NaCl eluate pool. Experimental conditions for the frontal analysis run are $u = 76$ cm/h, column length = 3.0 cm, residence time = 2.4 min. Lines in (a) are predictions based on Eqs. (6)–(8) including a kinetic resistance to binding with rate constants in Table 2.

a kinetic resistance to binding must be considered. This resistance has only a small effect on single-component breakthrough curves but deeply impacts the binary adsorption behavior. Thus, it must be included in the model in order to quantitatively predict binary breakthrough and hence the purity and yield attainable by frontal analysis. Comparison of model predictions with previously obtained batch uptake kinetics data generated by confocal microscopy confirm the validity of this conclusion. No attempt was made to model the binary breakthrough curves at low Na⁺ concentrations. Although these conditions did not lead to a practical separation, modeling this behavior is interesting from a fundamental viewpoint. One possibility is that our predictions for these conditions failed because the SMA isotherm is simply unable to describe the relevant binary adsorption equilibria. Another possibility is that our binding kinetics model is overly simplistic. Work is underway to independently measure the competitive binding equilibrium and kinetics with the goal of developing more accu-

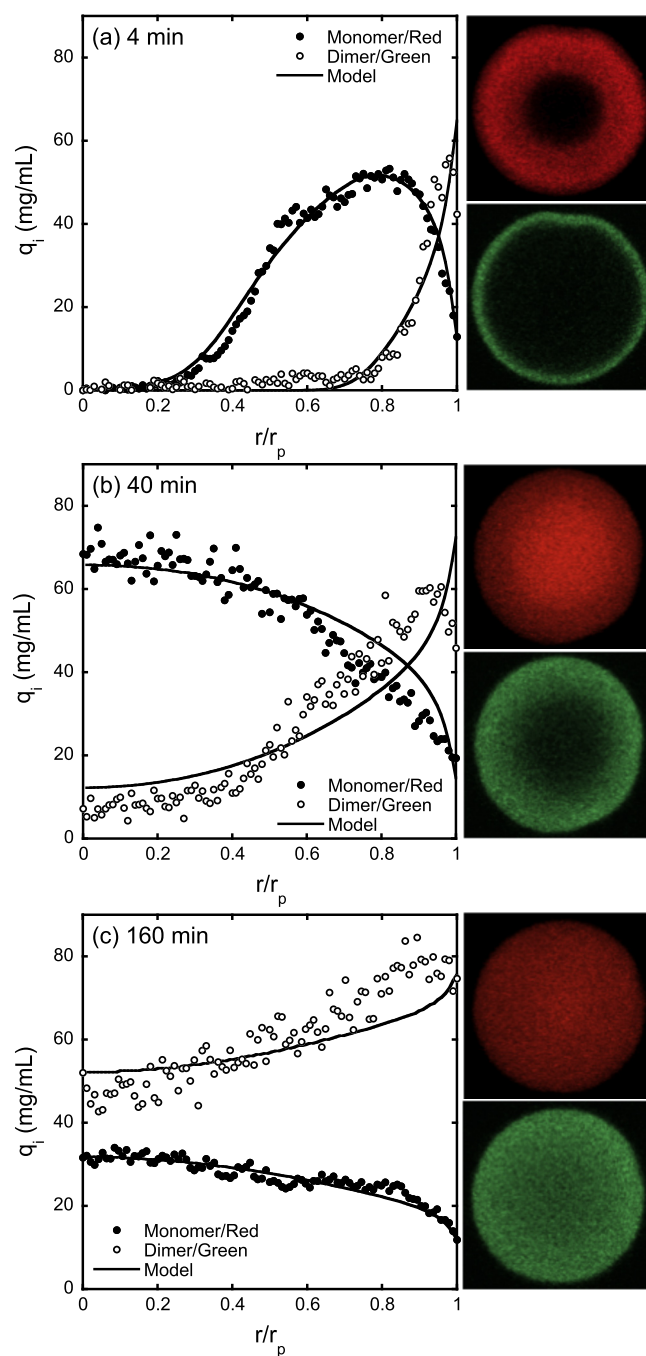


Fig. 6. Comparison of experimental intraparticle concentration profiles during co-adsorption of a 2 mg/mL monomer-dimer mixture containing 15% dimer in pH 7 buffer with 80 mM Na⁺ with model predictions based on Eqs. (7)–(8). Confocal microscopy images are from Ref. [10]. Solid and open circles are the normalized fluorescence intensity profiles of the monomer and dimer, respectively. Lines are model predictions without adjusting any of the parameters.

rate kinetic models that could explain the unexpected selectivity trends as a function of counterion concentration.

From a more practical viewpoint, our results show that frontal analysis can be developed as a useful tool to remove aggregates from mAb feedstock. Compared to ordinary bind-elute modalities, frontal analysis has the advantage of more fully utilizing the resin binding capacity and minimizing dilution of the purified product. Resin with relatively small particle size and with relatively large pores are needed for such a process in order to minimize band-broadening effects. Finally, though it is beyond the scope

of this work, alternative types of chromatographic media such as monoliths or membranes, where mass transfer is not rate limiting, may be well suited for frontal analysis. Predicting performance, in these cases, would however require a description of other band-broadening factors which affect these systems (film mass transfer and axial dispersion), in addition to the kinetic resistances associated with the exchange of bound protein molecules observed in this work.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.chroma.2017.04.014>.

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